

Image Enhancement of Dark, Blurred and Noisy Images

PCD Assignment 02 — Pengolahan Citra Digital (Digital Image Processing)

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1. Objective

The objective of this assignment is to enhance three real degraded photographs, one for each of three common image problems: a **dark** (low-light) image, a **blurred** image and a **noisy** image. One suitable enhancement method is implemented for each problem: **gamma correction**, **Richardson–Lucy deconvolution** and **non-local means denoising**, respectively. The results are then analysed. All source code is in the Jupyter notebook *PCD_Assignment02.ipynb*.

2. Test images

The three images are real photographs (not artificially degraded) taken from public image datasets hosted on GitHub (Table 1). The notebook downloads them from these sources and stores them unchanged in *images/input/*. Since no clean original exists for a real photograph, the results are evaluated visually and with **no-reference measures**: mean (brightness), standard deviation (contrast), Shannon entropy (information), variance of the Laplacian (sharpness) and the estimated noise standard deviation σ .

Table 1. Test images and their sources (full URLs are in the notebook and README).

Problem	Image	Source	Credit
Dark	LIME 2.bmp	LIME low-light dataset — Guo, Li & Ling, IEEE TIP 2017; file from github.com/Li-Chongyi/Zero-DCE	research dataset
Blurred	fishes.jpg	Real blurred images of Ren et al., “SelfDeblur”, CVPR 2020; github.com/csdwren/SelfDeblur	research test image
Noisy	RNI15 Flowers.png	Lebrun, Colom & Morel, “The Noise Clinic”, IPOL 2015; file from github.com/cszn/FFDNet	research test image

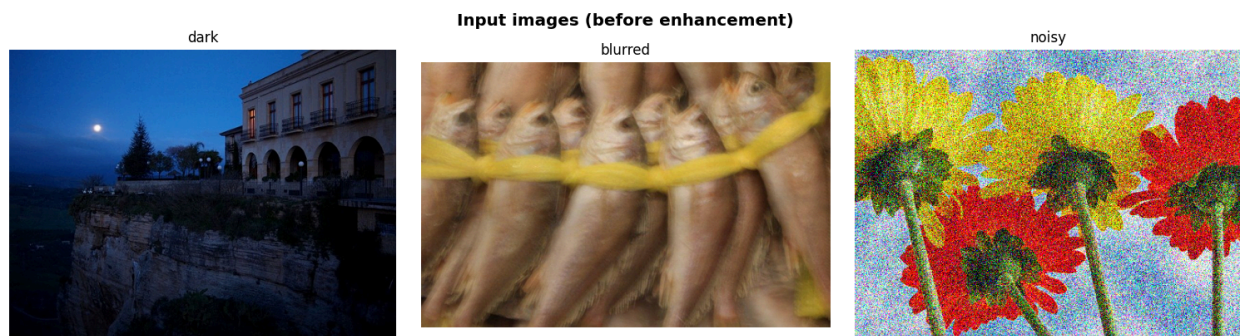


Figure 1. The three input images before enhancement.

3. Methods

3.1 Gamma correction (dark image)

Gamma correction is a point transformation $s = r^\gamma$ with r, s in $[0, 1]$. For $\gamma < 1$ the curve lies above the identity line, so dark values are raised strongly while bright values change little. It is applied only to the L^* (lightness) channel of the CIELAB colour space, so the colours are preserved. $\gamma = 0.5$ was used.

3.2 Richardson–Lucy deconvolution (blurred image)

A blurred image is modelled as $g = f * h + n$, where f is the sharp image, h the point-spread function (PSF) and n noise. Richardson–Lucy estimates f iteratively: $f^{(k+1)} = f^{(k)} \cdot [(g / (f^{(k)} * h)) * \tilde{h}]$, where $\tilde{h}$ is the flipped PSF. The true PSF of a real photograph is unknown, so a Gaussian PSF was assumed. After testing $\sigma = 1, 2$ and 3 , $\sigma = 2$ with 20 iterations was chosen.

3.3 Non-local means (noisy image)

Non-local means replaces each pixel by a weighted average of pixels whose surrounding *patch* is similar: $w(p, q) = \exp(-\|P(p) - P(q)\|^2 / h^2)$. Unlike mean or Gaussian filters, which average neighbouring pixels and blur edges, it averages similar structures, so noise is removed while edges are kept. The noise level σ is estimated from the image, and $h = 0.8\sigma$ with 5×5 patches is used.

4. Results and Analysis

Table 2. No-reference measures before and after enhancement (after = highlighted).

Image		Mean	Std	Entropy	Sharpness	Noise σ
Dark	before	0.152	0.131	6.590	350	0.0038
	after	0.330	0.185	7.259	838	0.0061
Blurred	before	0.450	0.152	7.250	37.798	0.0021
	after	0.450	0.155	7.294	26.591	0.0008
Noisy	before	0.541	0.274	7.931	45644	0.232
	after	0.540	0.196	7.061	149	0.0066

4.1 Dark image — gamma correction

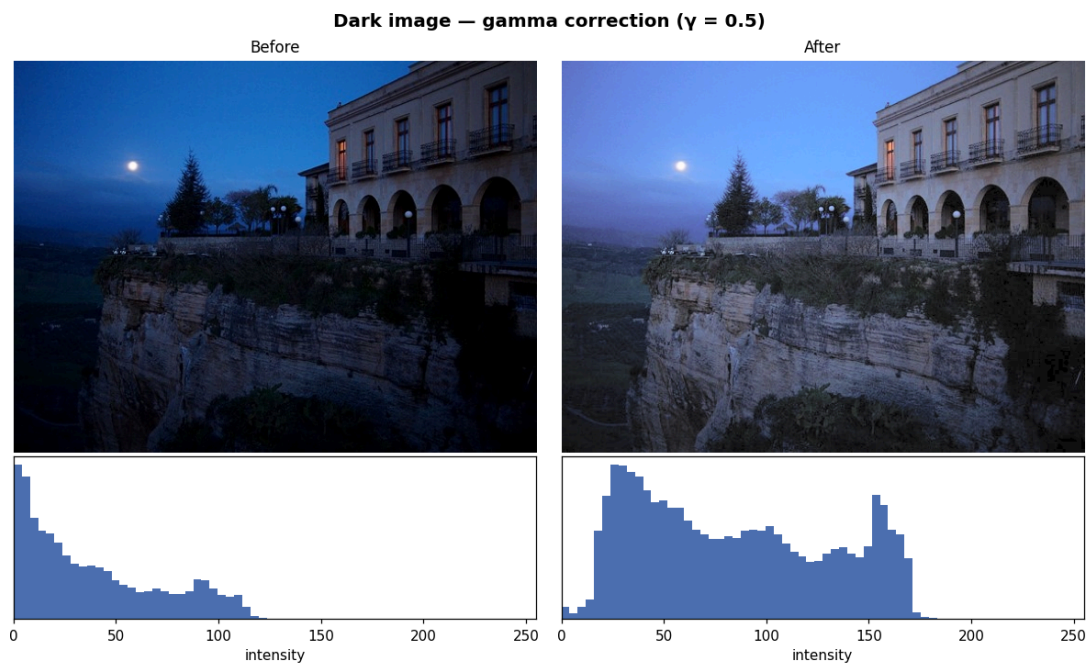


Figure 2. Dark image before and after gamma correction ($\gamma = 0.5$), with luminance histograms.

Gamma correction with $\gamma = 0.5$ reveals the cliff, the trees and the building facade, which were almost invisible in the input. The histogram, which was packed into the lowest intensities, is spread over a much wider range. The mean brightness more than doubles ($0.152 \rightarrow 0.330$), and contrast (std $0.131 \rightarrow 0.185$) and entropy ($6.59 \rightarrow 7.26$ bits) both increase, so more detail is visible. Because only L^* is changed, the colours of the scene (the blue evening sky, the warm window lights) are kept. There are two limitations. First, brightening also makes the hidden sensor noise more visible (estimated σ $0.0038 \rightarrow 0.0061$). Second, pixels that were completely black (value 0) stay black, since $0^\gamma = 0$, so a few small dark specks remain in the deepest shadows. A smaller γ would brighten more, but it would make the image look flat and washed out.

4.2 Blurred image — Richardson–Lucy deconvolution

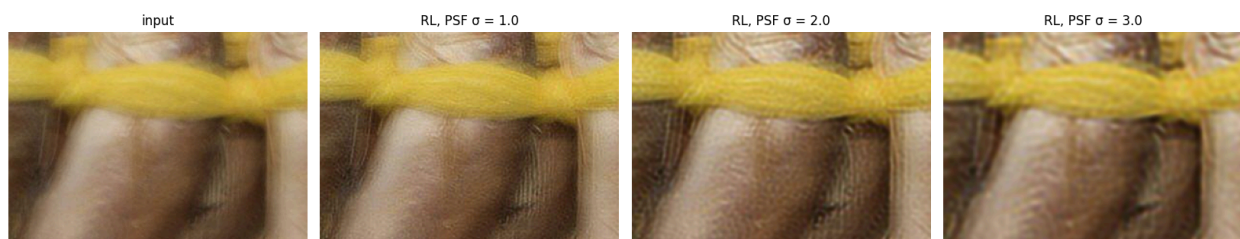


Figure 3. Choosing the assumed PSF width (close-up): $\sigma = 1$ too weak, $\sigma = 3$ gives ringing.



Figure 4. Blurred image (close-up) before and after Richardson–Lucy deconvolution ($\sigma = 2$).

Richardson–Lucy deconvolution clearly sharpens the image. In the close-up, the individual fibres of the yellow string and the edges and texture of the fish become visible, while the input is smooth and smeared. The PSF test shows why $\sigma = 2$ was chosen: $\sigma = 1$ changes very little, while $\sigma = 3$ is too large for the real blur and starts to create ringing (repeated edge patterns). Contrast and entropy increase slightly (std $0.152 \rightarrow 0.155$, entropy $7.25 \rightarrow 7.29$ bits). Interestingly, the Laplacian-variance sharpness value decreases ($38 \rightarrow 27$) and the estimated noise also falls ($0.0021 \rightarrow 0.0008$). RL restores the mid-frequency edges that were blurred, but it also suppresses pixel-level noise and JPEG texture, which is exactly what the Laplacian measures. A simple sharpness number therefore does not always agree with what the eye sees, and visual inspection is necessary. The main limitation of the method is that the PSF must be known or guessed; a wrong PSF gives artefacts instead of detail.

4.3 Noisy image — non-local means

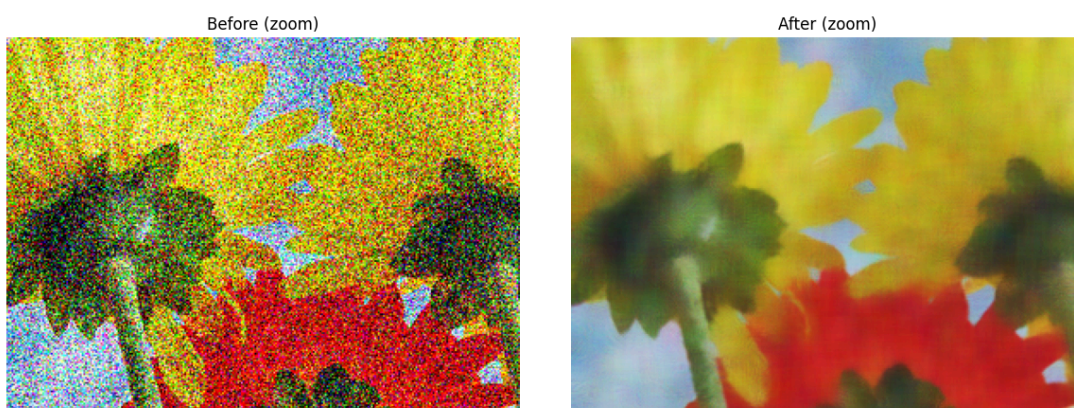


Figure 5. Noisy image (close-up) before and after non-local means denoising.

The input contains extremely strong real colour noise (estimated $\sigma \approx 0.23$ on a 0–1 scale). Non-local means removes almost all of it: the estimated noise falls about $35\times$ ($0.232 \rightarrow 0.0066$). The sky becomes smooth, and the flowers and stems keep clear outlines, because each pixel is averaged only with pixels whose neighbourhood looks similar. The Laplacian variance falls from 45 644 to 149, which shows how much of the original 'detail' was only noise. The standard deviation ($0.274 \rightarrow 0.196$) and entropy ($7.93 \rightarrow 7.06$ bits) also decrease, because the random noise had artificially increased both. The price is that very fine texture, such as the thin lines on the petals, is smoothed away, giving a slightly painted look. This is unavoidable at such a high noise level, where the fine details are buried under the noise.

5. Conclusion

Each type of degradation needs its own method:

- **Dark image $\rightarrow$ gamma correction.** A simple, fast point operation that brightens shadows much more than highlights. Applied to the L^* channel it preserves colour. Its drawbacks are that it also amplifies hidden noise and cannot recover pixels that are completely black.
- **Blurred image $\rightarrow$ Richardson–Lucy deconvolution.** Unlike simple sharpening filters, it models the blur and inverts it, so it recovers real edges and texture. It depends on the point-spread function, which for a real photo has to be estimated; a wrong PSF produces ringing.

- **Noisy image → non-local means.** It averages similar patches instead of neighbouring pixels, so it removes very strong noise while keeping edges. Very fine texture is lost when the noise is heavy.

For real photographs no clean reference exists, so the results were judged visually and with no-reference measures (brightness, contrast, entropy, sharpness, estimated noise). These measures are useful, but they can disagree with visual quality: the deblurred image, for example, has a lower Laplacian variance although it looks sharper.